

Jah'dir Cartagena-Rivers

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EDUCATION

Temple University

B.S. Mechanical Engineering | **GPA: 3.80**

Expected May 2030

Lankenau Environmental High School

High School Diploma | Salutatorian | **GPA: 4.05**

2022 – 2026

EXPERIENCE

Nanomaterials Research Intern

Jun 2024 – Aug 2024

West Chester University

- Synthesized fluorescent gold nanoclusters capped with amino acids, proteins, and polymers, assessing cellular interactions and optical properties using circular dichroism and UV-vis spectroscopy
- Investigated fluorescence quenching kinetics for selective heavy metal ion detection, establishing sensing thresholds for Copper (**Cu²⁺**) and Lead (**Pb²⁺**) in aqueous solutions

Aquaculture & Environmental Education Intern

Aug 2025 – Jun 2026

Philadelphia Water Department

- Designed multi-year experiments on freshwater mussel survivability under projected temperature increases, managing live specimens and conducting water quality tests toward publishable findings
- Tracked shell dimensions and mass across **20+ tagged specimens** over a **16-week study**, recording **130+ precision data points** to analyze growth rates and tank survivability

Environmental Microbiology & Public Health Intern

Aug 2025 – Jun 2026

Lankenau Environmental High School

- Collected and analyzed **nearly 100 microbial swabs** across SEPTA subway lines and bus routes to track bacterial presence patterns; presented findings at the Philadelphia Service Awards
- Applied standardized agar plating, positive and negative controls, and a **0-to-4 colony density scale** to generate environmental health data

Policy and Constituent Services Intern

Apr 2026 – May 2026

City Hall, Office of Council Member Nina Ahmad

- Conducted policy research and facilitated communication between the council office and Philadelphia residents, supporting constituent outreach and administrative operations

Community Barbershop Outreach Volunteer

Jun 2025 – Aug 2025

Independent Volunteering

- Logged **80+ volunteer hours** providing free grooming services to shelter residents and coordinating weekly event logistics

PROJECTS

Computer Vision Media Pipeline | Lead Developer / ML Engineer

Aug 2025 – Present

- Architected an automated media classification pipeline in Python using PyTorch, CLIP embeddings, and OpenCV frame extraction.
- Implemented weighted three-signal confidence fusion combining CLIP visual cosine similarity (**30%**), OCR watermark extraction (**40%**), and fuzzy string matching (**30%**); set a **0.90 decision boundary** for automatic routing and 50–89% for review.
- Built FastAPI and Streamlit dashboards with real-time logs, manual triage, and Plex REST API integration.

Automated Application Pipeline Engine & Browser Copilot | Creator and Lead Developer

July 2026 – Present

- Built a full-stack automation platform in Python and React using Playwright to navigate multi-step job applications on LinkedIn, Indeed, and Handshake with stealth anti-detection
- Implemented an automated IMAP email parser and WebSocket pipeline to track interview invites and application statuses on a real-time Kanban board

Gyromi | Creator and Lead Developer

May 2026 – Present

- Engineered an open-source, cross-platform multimedia hub in Flutter, Dart, and Rust for Windows and Android with offline-first sync via Google Drive REST API and **PBKDF2-HMAC-SHA256** credential hashing.

TECHNICAL SKILLS

Languages: Python, Dart, Rust, Java

Frameworks and Libraries: PyTorch, Hugging Face Transformers, OpenCV, FastAPI, Flutter, Playwright, React, WebSockets

Other: laboratory techniques, data analysis, GIS, Microsoft Office Suite

Certification: OSHA 10 General Industry